Reflection

Camille: During this lab, I learned how K-Means clustering identifies patterns by organizing similar data points into groups without relying on predefined labels. I developed a better understanding of how the number of clusters affects the model and that setting n_clusters=4 instructs it to form four groups. I learned that cluster labels indicate which group each data point belongs to, while cluster centers represent the average position of the points assigned to each group. The plotting code helped me understand how colors and markers can make clustering results easier to interpret. I also learned that Python treats lines beginning with # as comments, meaning those lines provide instructions or explanations but do not execute. Encountering the error that KMeans was not defined taught me that a notebook depends on running its setup cells in the correct order. This experience showed me that troubleshooting involves reading the error message and checking which earlier steps the code requires. Although I still need practice, I gained a clearer understanding of how the model, cluster assignments, and visualization work together.

Min:

This lab made me learn the differences between supervised and unsupervised machine learning techniques. In previous labs, we always had a specific target to predict; however, here we only had input features and had to let the algorithm find patterns on its own. During the K-Means analysis, I noticed that the algorithm cannot itself determine the optimal number of clusters, so we applied the elbow method for that purpose. The value of inertia was reduced from about 19,780 for k=1 to 362 for k=4 and barely changed after that, so we chose k=4 as the number of clusters. I realized that choosing the highest k value isn't a good idea, since inertia always decreases as k increases, and with too many clusters, real groups get broken into meaningless pieces. The PCA part helped me understand how to reduce the dimensionality of the data without losing too much information: we reduced the Iris data from 4 features to 2 and still kept 97.77% of the variance. The most interesting aspect was that PCA was not showing the species labels, but setosa was still able to completely separate from the other two. What I learned from this lab is how difficult it is to evaluate results without labels and our data was made up with 4 clear groups, but in reality, data can be messy and ambiguous.

Chris: In this lab we learn how to use unsupervised learning to discover structure in a large amount of data. We apply in particular Principal Component Analysis (PCA) to the Iris data set. The main goal of this lab was to reduce the Iris data set from 4 dimensions to 2, and to understand how much of the information is actually kept. We find that two principal components keep 97.77% of the original information, and we can thus simplify the data set considerably while still keeping most of the information.I was able to see how dimensionality reduction could make it easier to plot and analyze data. Using the appropriate modules in Python (Scikit-learn, Pandas, and Seaborn) I was able to create plots of the data after it had been transformed. I was also able to see how PCA and K-

Means, two forms of unsupervised learning, could be used to make a machine learning workflow more efficient.

Rigoberto:

This lab helped me understand unsupervised learning and how it is different from supervised learning. Before working through this assignment, I knew that machine learning models could identify patterns and make predictions, but I did not fully understand how a model could find patterns without being given the correct answers. After going through the notebook, I have a better understanding of how unsupervised learning works and why it is useful when working with data that does not have labels.One of the concepts I learned was the difference between supervised and unsupervised learning. Supervised learning uses labeled data, meaning the model is given input features and the correct answers during training. Unsupervised learning does not have those predefined answers. Instead, the model analyzes the data and looks for similarities, patterns, and relationships. This was also the main concept I focused on when completing Question 1 in the Knowledge Check. Understanding this difference helped me see that machine learning is not always about predicting something. Sometimes the goal is to discover information that was not obvious before.Another important part of the lab was K-Means clustering. I learned that K-Means organizes data into groups based on similarities between data points. It uses centroids to determine where each cluster is located and repeatedly adjusts them until the groups become more stable. I also learned about the Elbow Method, which helps determine how many clusters should be used. Instead of automatically choosing the largest number of clusters, we look at the inertia values and identify where adding more clusters stops making a significant improvement.I also learned about Principal Component Analysis, or PCA. At first, I found dimensionality reduction a little confusing because I thought having more features would always be better. However, I learned that too many dimensions can make data more difficult to analyze and visualize. PCA helps reduce the number of features while preserving as much important information as possible. This showed me how machine learning techniques can make complicated datasets easier to understand.One thing I took away from this lab is that every step in a machine learning workflow has a purpose. Choosing the number of clusters, understanding the data, and reducing dimensions are all important decisions that affect how we interpret the results. I also realized that different algorithms are designed for different situations, so understanding the problem is important before choosing a method.Overall, this lab improved my understanding of unsupervised learning and gave me a better idea of how it can be used in real-world situations. For example, businesses could use clustering to group customers based on similar behaviors or purchasing habits without already knowing which groups those customers belong to. I feel more comfortable explaining the difference between supervised and unsupervised learning now, and I understand better how techniques like K-Means and PCA help us discover useful information from data. This lab also helped me connect what I learned in previous machine learning activities to a different approach to training and analyzing models.